Project Report on

Bharat AI-SoC Student Challenge

Problem Statement:

Offline, Privacy-Preserving Hindi Voice Assistant on Raspberry Pi

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I) Abstract

This Project presents a fully offline Hindi voice assistant **Krish** on Raspberry Pi that works fully offline, ensuring that the privacy of the user is preserved. Krish uses Vosk ASR to recognize Hindi speech and Piper TTS to provide Hindi speech as output. In its current state, Krish handles setting alarms, fetching time, playing music, and providing emergency advice and numbers without cloud dependency. Krish runs on CPU with low latency (~1.7 secs).

II) Objective

- To design and deploy a fully offline Hindi voice assistant on Raspberry Pi, ensuring complete user privacy.
- To process Hindi voice commands locally using on-device ASR and TTS, without reliance on cloud services or internet connectivity.
- To achieve low-latency operation with an average response time below 2 seconds for common user queries.
- To support predefined local queries such as time, alarms, music playback, and emergency assistance through accurate intent recognition.
- To validate system performance on CPU-only execution, measuring latency, accuracy, and resource utilization.

III) System Architecture

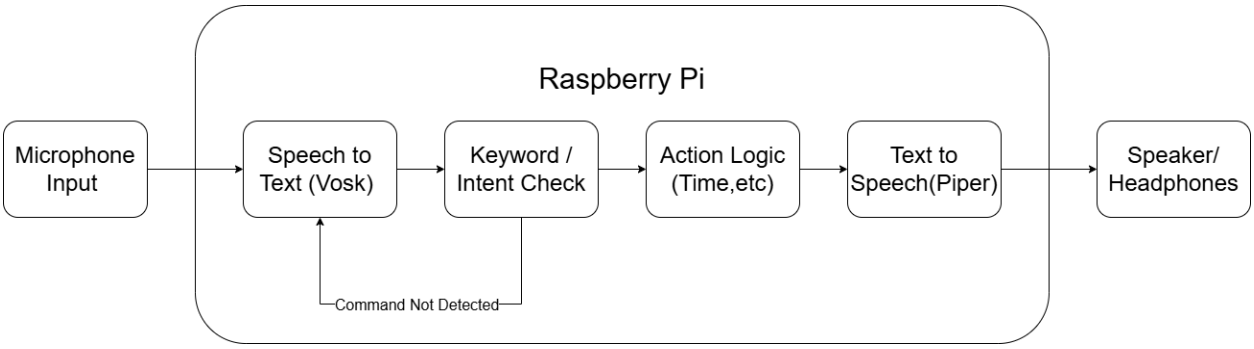

fig1. Simple Block Diagram

System flow:

Microphone → ASR (Vosk) → Intent Detection → Command Execution → TTS (Piper) → Speaker

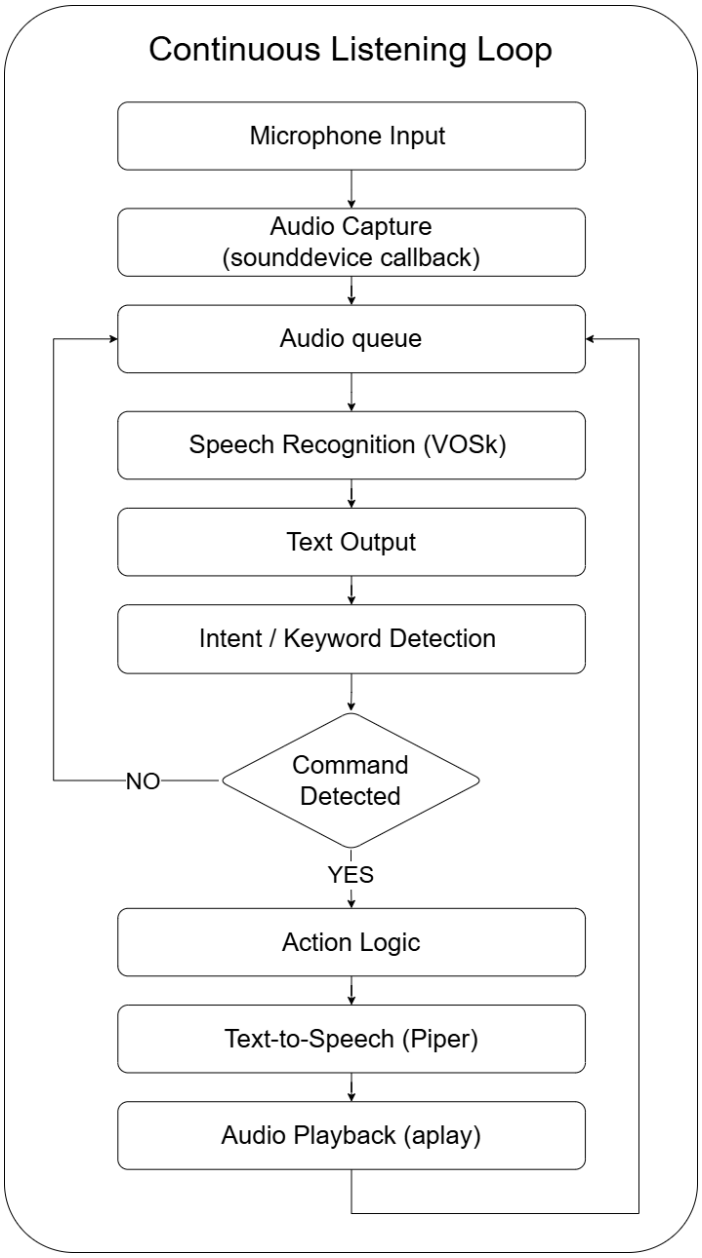

Fig2. Flow Chart

Steps:

1. Capture audio from microphone
2. Convert speech → text using Vosk
3. Detect intent using Hindi keywords
4. Execute command (alarm/music/time/etc.)
5. Speak response using Piper TTS

IV) Hardware Used

Component	Specification	Purpose
Raspberry Pi 4	ARM Cortex –A, 4GB	Main processing
USB Microphone	18 KHz frequency response, 60dB SNR	Speech Input
Speaker	40W Stereo Sound, 105 dB Output	Voice Output

System is designed to run fully on CPU without any accelerator to maintain low costs and to maintain user privacy

V) Software Used

- Python
- Vosk (Offline Hindi ASR)
  - Word Error Rate (WER): ~8 – 18%
  - Sentence Recognition Accuracy: ~82 – 92%
- Piper TTS (Offline speech output)
  - MOS (Mean Opinion Score): ~3.6 – 4.2 / 5
  - Provides Hindi Voice support
  - Computationally very light
  - Low latency
- sounddevice (Audio input)
- threading + subprocess

VI) Methodology

1. Continuous audio captured using callback stream
2. Vosk offline model converts speech to text
3. Rule-based keyword detection used for intent recognition
4. Commands supported: time, alarm, music, emergency, math
5. Piper TTS generates Hindi speech output offline

VII) Optimization Techniques

We have applied various methods for optimization:

1. Vosk which is a lightweight Hindi small model, is used to reduce CPU utilization.
2. Rule based intent recognition instead of heavy NLP to reduce the response time for a better user experience.
3. Callback audio streaming for non-blocking audio capture.
4. Threading is used to implement a non-blocking execution of commands.

VIII) Commands

Keyword	Action
"नाम", "कौन", "आप कौन"	Speaks his name
"समय","टाइम","वक्त","बजा"	Tells the time
If "गाना" and चलाओ"or"लगाओ"or"शुरू"	Plays a song
If "गाना" and "बंद"or"रोक"	Stops the song
If "अलार्म" and "लगाओ"or"सेट"or"कर"	Sets alarm
if "अलार्म" and "बंद"or"रोक"or"स्टॉप"	Stops alarm
"प्लस", "गुणा","भाग",etc	Does the calculation
"गैस","भूकंप","आग","करंट",etc	Gives emergency protocol
If any Indian State	Tells the capital city
"मुहावरा"	Says an Idiom
"दोहे", "dohe","दोहा"	Tells a Doha
"व्याकरण", "संज्ञा", "क्रिया", "काल"	Explains the grammar
"वैज्ञानिक", "कलाम", "रमन"	Gives Information of Scientists
"योजना", "सरकारी"	Explains a government scheme
"जोक", "joke","चुटकुला"	Cracks a Joke
"कहानी", "story"	Tells a Story
"दोहराओ"	Repeats
"बस करो"	Stops

IX) Performance

Metric	**Result**
Avg Response Time	~1.7 secs
Offline Operation	Yes
Commands Working	Yes
CPU Usage	60%-75%
Accuracy	85%-90%

X) Challenges

Here are some challenges we faced during the execution of this project:

1. Issue with the Hindi voice Recognition
2. Implementing the accurate intent matching in Hindi text
3. Noise affecting the quality of voice recognition
4. Running speech pipeline on limited CPU
5. Parsing Hindi time expressions was complex

XI) Results

We have achieved the following:

1. The assistant works fully offline
2. Responds Correctly to Hindi voice commands
3. Real-time voice response achieved
4. A working set of commands including alarm, music, time and emergency help

XII) Future Improvements

We would want to add these in future:

- On-device learning
- Improve Hindi ASR accuracy
- Add real SOS GSM calling
- Support more commands

XIII) Conclusion

This project achieves the stated objectives. The work covered planning, building and validating the results to successfully build Krish as an offline Hindi voice assistant, conserving users' privacy with low latency.